

Trust, but Verify

Accounting and Finance Research in the Age of AI

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PART ONE

The problem we always had

Delegating work you cannot directly watch is the oldest problem in the economics of organization. AI changed what the pieces of it cost.

Delegation was always the problem

Trust is delegation while verification is monitoring. Coase, Alchian and Demsetz, Jensen and Meckling — the economics of delegating work you cannot directly observe was established fifty years ago.

AI does not change the problem, rather it changes the prices.

	YOUR RA	THE MODEL
Speed	A semester	Minutes
Scale	One task at a time	The whole sample at once
Opacity	You can read her spreadsheet and see how she coded the ambiguous cases	You get a plausible reconstruction, not a log
Stakes	She posts a bond: a reputation, a job market, letters I have to write	LLM posts no bond, if LLM hallucinates no consequence to the machine

Jensen and Meckling priced this in 1976. Only the cost of production has moved.

Verification debt

Every research task has two costs: making it and checking it. You used to get the checking almost free, because building the thing was what taught you where it was weak. **Delegate the building and you give that up.**

cost of a claim = p (produce) + v (verify)

Delegating lowers p and raises v. It is worth doing only if p falls by more than v rises.

	CHEAP TO MAKE	COSTLY TO MAKE
COSTLY TO VERIFY	THE DANGER CORNER AI-coded text measures. AI-drafted regression tables.	HIGH STAKES BOTH WAYS A novel identification strategy. A new estimator.
CHEAP TO VERIFY	ROUTINE Variable-definition tables. Table formatting.	COSTLY BUT CHECKABLE A hand-coded audit sample. Entry against a known source.

Checking your own work catches slips, not bad assumptions.

We need to think of what the standard is

What settles it is not how hard the task is. It is whether anything outside the work or objective can tell you the answer is right.

DELEGATE IT — THE STANDARD IS OUTSIDE THE WORK

KEEP IT — THE STANDARD IS IN YOUR HEAD

Does this citation exist?

Does this variable measure the construct?

Did the merge keep the rows it should?

Is the effective date the one that binds?

Does the code pass the test?

Is this the right control group?

Do the row counts tie to the source?

What evidence would change my mind?

Delegate to the extent there is an external standard. Don't delegate the construct.

The bottleneck in research has moved to checking

Reading a predoc's code still costs what it always did. However, **LLMs now generate in an afternoon what used to take a semester.**

IF PRODUCTION GETS	THE WORK GETS	VERIFICATION WAS	VERIFICATION IS NOW
2× cheaper	1.5× faster	30% of the cost	46%
10× cheaper	2.7× faster	30% of the cost	81%
100× cheaper	3.3× faster	30% of the cost	98%

Amdahl's law, and it does not care how good the model gets. Illustrative, with verification at 30 percent of unit cost; the ceiling is 3.3×. $\bar{v}$ is not fixed by nature — batch it, sample it the way an auditor would, or write the test once and let it check forever. None of that happens by default.

Production 100× cheaper moves verification from 30 percent of the cost to 98 percent.

The flashy machine

Checking is now the expensive part. We tend not to spend more on it, because the output does not look like it needs checking. Readers used to judge care by how finished the work looked as polishing a paper took time a careless author would not spend, and now LLMs now does it for both.

WHAT REPLACES IT

Being easy to check still separates careful work: **cheap if you are right, expensive if you are wrong.**

- A prediction stated before the result
- Code and data a stranger can run
- The condition under which you would be wrong
- An unscripted answer to “why did you do it that way?”

SIXTEEN EXPERIENCED DEVELOPERS, 246 TASKS

19% slower

WHAT ACTUALLY HAPPENED

24% faster

WHAT THEY FORECAST

20% faster

WHAT THEY BELIEVED AFTERWARD

BECKER, RUSH, BARNES AND REIN (2025), METR • ARXIV 2507.09089 • 16 DEVELOPERS, 246 TASKS, REPOS THEY HAD WORKED FOR YEARS

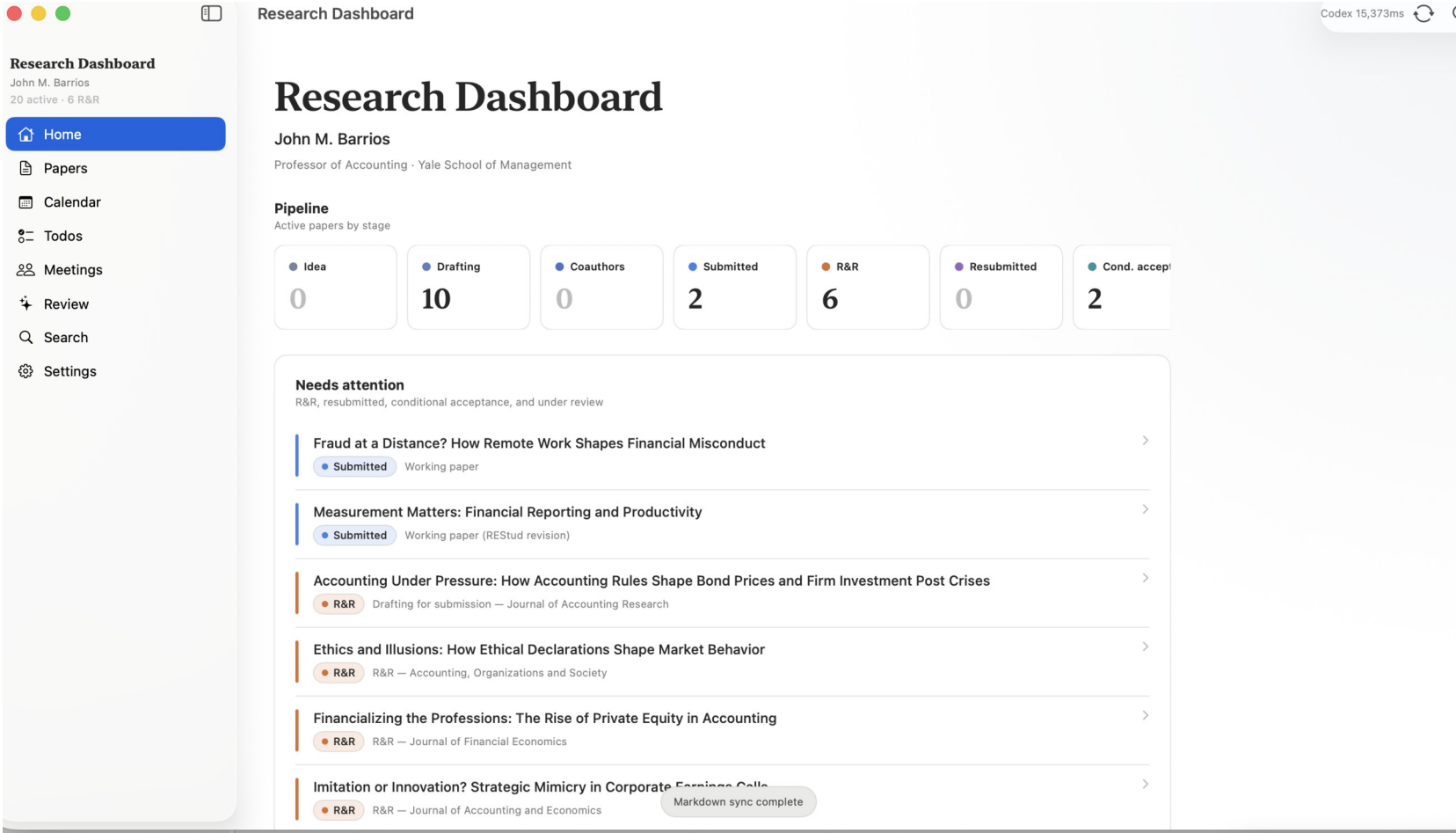
Looking right is not evidence that it is right.

PART TWO

How I use LLMS in my work

A research dashboard, an agent, and a live connection to WRDS and where I draw the line with delegating.

The research dashboard



What the dashboard does

Every paper I have going is a card. Coauthors, stage, the journal it's at, deadlines, and the last time anything in the folder moved— synced from markdown files any coauthor could read.

THE MORNING BRIEF

Reads the whole corpus and tells me what moved and what is due or likely to be late.

OPEN IN AN AGENT

One command to open any paper in a coding agent, in that paper's folder, with that paper's context.

SEMANTIC SEARCH

Finds the argument I know I made three years ago and cannot name.

Everything the agent proposes lands in a suggestion inbox, and nothing is written until I confirm it.

Decision management delegated, decision control retained — Fama and Jensen (1983).

The agent I work with, daily



Hermes is my messenger and personal assistant. Papers, revisions, teaching mail, Stata logs, and the next thing I said I would do. He keeps all papers in his head. Treat him like staff rather than a search box — and staff get scope, instructions, and review.

	WHAT HE DOES	WHAT HE WILL NOT DO
Scope	Reads and writes code, pulls from WRDS, drafts, checks citations against a database he does not control	Choose the construct, set the control group, or decide what the paper claims
The catalog	Opens the YAML, then the path. Knows which manuscript is conditionally accepted and which folder is the file	Invent a DOI, an award, or a status to make the catalog look tidier — a gap in a catalog is information
Empirical work	Sends Claude Code into a git worktree, credentials kept out, and returns a job id, a branch, and a diff	Pretend to be the estimator

Every empirical run comes back as a job id, a branch, and a diff I have to read.

The WRDS connector – streamlined data collection

A question, a schema lookup, a query, a panel — one conversation, against my own WRDS account.

YOU ASK

“What does crsp.msdelist look like, and how many delisting events are missing a return?”

IT LOOKS BEFORE IT WRITES

wrds_describe_table → crsp.msdelist · columns, types, row count

IT RUNS THE QUERY

wrds_run_sql → 29,000 events grouped by delisting code, mean dlret, count missing · 4 seconds

WHAT TO WATCH

- 1 It reads the schema first.** before it writes a line of SQL.
- 2 The SQL is in the transcript.** so you can re-run it, and so can a referee.
- 3 Four seconds.** which is the whole problem. Nothing here checked whether the panel is right.

WRDS-MCP · CRSP.MSELIST · ONE CONVERSATION, FOUR SECONDS

What the connector does not check

Compustat, CRSP, Form 4, TRACE, TAQ — schema exploration, SQL, download, in the conversation.

Delisting returns in CRSP, and where they go missing

DELISTING REASON	EVENTS	MEAN DELISTING RETURN	RETURNS MISSING
200s merger / acquisition	14,494	+0.0262	17 (0.12%)
300s exchange	1,067	+0.0242	1 (0.09%)
400s liquidation	2,682	+0.0148	8 (0.30%)
500s dropped (for cause)	10,860	-0.2316	164 (1.51%)

WRDS-MCP • CRSP.MSEDELIST • 29,103 EVENTS • LIVE QUERY 0

164 of the 190 missing delisting returns — 86 per cent — sit in the one category that averages -23%.

Drop the rows where dlret is missing and you have silently deleted the disasters.

Schema lookup, query writing and download are now largely automated in my workflow. Deciding which check would settle the claim is not.

PART THREE

AI and the production of knowledge

What cheap generation does to the stock of things we actually know.

Candidate output vs credible knowledge

N = f (raw information, AI processing, computation, prompts, search)
nearly every one of these just got much cheaper

K = g (N, screening, verification, research design,
institutional knowledge, judgment)
these did not — or not equally

Privately, you are paid for claims that **look** credible. Socially, we gain only from claims that **are** —
net of what they cost everyone else to check.

Verification needs a standard: a construct, an estimand, a ground truth and the LLM cannot supply the standard.

Screening and the verification burden: marginal-candidate problem

Search theory says cheaper draws should make you *pickier*, true, when you can see quality. We rank on plausibility and end up paying for truth, so what grows with the pool is the gap between how good the winner *looks* and how good it *is*. **We would discount our best idea by how many we looked at.**

SCREENING **You do this before you spend anything, and none of it requires running code**

- Is the question economically important, or only empirically available?
- Is there a credible source of identifying variation, or am I hoping controls do the work?
- Is the construct observable with tolerable error?
- Would this change an expert's belief if true — and would anyone notice if it were false?

$$\text{Verification burden} = \frac{\text{candidate volume} \uparrow \times \text{claim complexity} \uparrow \times \text{verification intensity}}{\text{verification productivity} \uparrow}$$

Generation tools recover a large fixed cost over hundreds of millions of users. Verification tools have a few thousand, and the benefit accrues to readers.

Two kinds of verifications: task checks and claim checks

TASK CHECKS

Was the step performed correctly?

The source exists

The code runs

The table reproduces

The label matches a human benchmark

→ *AI is already good at this, and getting better*

CLAIM CHECKS

Does the evidence support the assertion?

The variable measures the construct

The comparison identifies the estimand

The timing matches when information arrived

The interpretation fits how the institution works

→ *AI is not good at this — and the reason is not model quality*

A better model can gather more evidence for a claim. **It cannot choose the construct, make an identifying assumption true, or put its name on the result.**

PART FOUR

AI and learning

Who keeps the skill once the tool is taken away.

What knowledge is left after the tool goes away

Three months after the tool was taken away, the advantage that survived sat, on average, entirely with the **senior lawyers**. 133 practising patent lawyers across 11 firms, pre-registered, blind-graded by expert attorneys.

WHILE THEY HAD THE TOOL

+0.34 SD

DRAFT QUALITY AT TEN DAYS • +0.38 SD AT NINETY

The juniors gained more while using it — the novice-skew result the field has repeated for three years.

THREE MONTHS LATER, THE TOOL REMOVED

+0.32 SD

POOLED, UNASSISTED • $P = .04$ • SENIORS +0.45 SD ($P = .02$), JUNIORS ~0

The juniors kept nothing on average. They **split** — fewer mediocre drafts, more poor ones and more good ones.

AUTOR, RODCHENKO, MARTIN, ISCENKO, STRAND, PEARL AND FERERE (2026), NBER 35720 • UNPUBLISHED; A SUBGROUP RESULT

The largest gains went to the most experienced lawyers. Implications for doctoral students and equity in the profession.

PART FIVE

What a paper is worth

Private rents, social cost, and who pays for the checking.

The paper and the human capital

Prior to LLMS, you could not produce a good paper without accumulating the capital. That is why a paper worked as evidence about an academic's quality. Agentic AI unbundles them.

THE ARTIFACT — THE PAPER	THE CAPITAL — THE RESEARCHER
Transferable, citable, cheap to copy once it exists	Built only by doing the work
Now produced at a pace no amount of your own understanding could have sustained	The thing a doctoral programme exists to produce
What the market pays for	What makes you a cheap verifier fifteen years later
Macnamara and coauthors argue that AI assistance targets precisely the demanding sub-tasks through which expertise gets built, and that skill decay proceeds without the performer noticing. Their evidence is the autopilot literature, not AI.	

The paper no longer certifies the researcher who produced it.

Do we get paid for research that got cheaper?

The scarce inputs are tenure lines, journal pages, and the top of a search committee's stack. Not papers. In a rank-order tournament the prize spread, not the level of output, sets the return to effort.

WHAT I WILL NOT OVERSELL

Whether we all work *more* is genuinely ambiguous. It turns on whether the tools substitute for effort or complement it, and on whether they leave us easier or harder to tell apart.

WHAT I CAN SIGN

The incidence. The prize is fixed by university budgets, so the marginal assistant professor does not capture the gain. Where the slot is priced — the tenure line — the employer takes it. **Where it is not — the journal page — it is burned in effort.**

One uncomfortable note. The people who do best here are the ones whose reputation already substitutes for verification, because a reader who cannot check falls back on the reputation of who is asking. Most of us in this room are collecting that rent. The doctoral students are paying it.

Ricardo where the slot can be priced, Tullock where it cannot.

Verification in the marketplace of ideas: the commons

I generate in ten minutes what costs you forty hours to evaluate, and I do not pay the forty hours. **As my private cost goes to zero and yours does not, the entire social cost of my marginal paper is the part I do not pay.**

THREE RESULTS, NONE OF THEM MINE

Congestion	Fewer downloads, worse outcomes — not only for the weak (Lusher, Yang and Carrell)
The baseline	Falling for decades even as effort grows — what this lands on (Bloom et al)
The referees	7–17% of peer-review text at four major conferences could have been substantially model-written — highest where reviewer confidence was lowest

LUSHER, YANG AND CARRELL (2023) JPUBE • BLOOM, JONES, VAN REENEN AND WEBB (2020) AER • LIANG ET AL. (2024) ICML

Referee hours are a flow; the credibility of the literature is a stock. Crowding you out of a slot is a transfer **if the papers are equally good** and a loss to the extent we rank on plausibility.

The verification machinery is itself being automated.

A human paper or a machine

Provenance matters exactly as much as verification is incomplete. Which is most of the time, because nobody can check everything.

THE POSITIVIST POSITION

A claim is worth what its content is worth. If the finding is true and it survived checking, provenance is a fact about production, not about the world.

THE BOND

An author is a residual claimant on the claim being true — somebody who loses something if it is wrong. **A bond economizes on checking.**

The cost neither position prices. Across 41 million papers, scientists using these tools publish three times as much and collect nearly five times the citations, while the range of topics contracts about five percent and scientist-to-scientist engagement falls twenty-two percent. Adoption is not random, so read these as association. Recombination produces real novelty; the problem is that our draws are now correlated.

HAO, XU, LI AND EVANS (2026), NATURE 649 • DOSHI AND HAUSER (2024), SCIENCE ADVANCES 10(28)

Correlated at one half, a thousand of us are worth fifty-seven independent draws.

What remains scarce

Our PhD is also a credential, and a credential is a measure. If it measures execution and calls that judgment, **readers will lean on it harder at exactly the moment it is telling them less.**

WHAT REMAINS SCARCE

Taste in questions
Institutional knowledge
Construct definition
Screening discipline
Validation design
Identification
Economic interpretation

Claim ownership

Somebody has to say, in their own name, ***I have looked at this and I believe it.*** A signature is worth something only because the signer can lose something, and a model has nothing to lose. But it is worth only as much as the chance that anyone checks — which makes every remedy on these slides a complement to verification capacity, not a substitute for it.

Producing answers is cheap now. Putting your name on one is not.

Thank you

Producing answers is cheap now. Putting your name on one is not.

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